

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

Design a Webpage using HTML and CSS

2. AIM

To create an attractive and responsive static webpage using HTML for page structure and external CSS for styling, layout, typography, and interactive components.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the separation of concerns between document structure (HTML) and visual presentation (CSS).
 - To implement external stylesheet linking using the `<link rel="stylesheet">` tag.
 - To apply universal reset styling and configure box-sizing (box-sizing: border-box).
 - To design navigation bars, hero welcome banners, and card layouts using CSS Flexbox (display: flex).
 - To implement interactive state styles using pseudo-classes such as `:hover` and `:focus`.
 - To style user input forms with modern padding, custom border styling, and button transitions.
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4. THEORY

CSS (Cascading Style Sheets) is a stylesheet language used for describing the presentation of a document written in HTML. CSS governs layout, colors, typography, spacing, and responsive behavior across diverse display dimensions. External CSS provides the highest level of maintainability by decoupling style definitions from HTML markup.

Theory of CSS Components and Layout:

- **CSS Box Model:** Every element is rendered as a rectangular box consisting of content, padding, border, and margin. Setting box-sizing: border-box ensures padding and border are included within the element's total width and height.
- **Flexbox Layout Model:** A one-dimensional layout model (display: flex) providing efficient alignment, distribution of space, and wrapping (flex-wrap: wrap) among items in a container.
- **Typography & Hierarchy:** Establishing clean readable typography using font families, line-height, and rem/em units.
- **Pseudo-Classes:** Modifying an element's style dynamically based on user interaction, such as `:hover` on links and buttons, or `:focus` on form inputs.

- **Card Containers:** Modular UI cards with background colors, subtle drop shadows (box-shadow), and borders to encapsulate content.
 - **Form Styling:** Transitioning away from default browser controls to modern, padded, accessible input fields.
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5. ALGORITHM / PROCEDURE

1. Create project directories: Record/Experiment - 2(week -2)/Src and Documentation.
 2. Create index.html and style.css in the Src directory.
 3. In index.html, establish HTML5 document structure and link style.css via <link rel="stylesheet" href="style.css">.
 4. In style.css, apply universal reset (* { margin: 0; padding: 0; box-sizing: border-box; }) and set base font properties on body.
 5. Construct a top navigation header with logo and flexbox navigation links.
 6. Design a hero banner with bold headline, descriptive lead text, and a stylized call-to-action button.
 7. Create content sections for About Me and Hobbies, using flexbox card layout for hobbies (Coding, Reading, Gaming).
 8. Style a contact form with vertical flex alignment, padded input controls, and custom focus indicators.
 9. Add a dark-themed footer with centered copyright text.
 10. Open index.html in Google Chrome and verify all styling rules, hover states, and responsive wrapping.
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6. PROGRAM

index.html (Lines 1-77)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My First Webpage</title>
  <!-- Link to external CSS stylesheet -->
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <!-- Top Navigation Header -->
  <header>
    <div class="logo">MyWebsite</div>
    <nav>
      <a href="#about">About</a>
      <a href="#hobbies">Hobbies</a>
      <a href="#contact">Contact</a>
    </nav>
  </header>

  <!-- Hero Welcome Banner -->
  <section class="hero">
    <h1>Welcome to My Website!</h1>
    <p>This is a simple static webpage created using basic HTML and CSS.</p>
    <a href="#about" class="btn">Learn More</a>
  </section>

  <!-- About Section -->
  <section id="about" class="content-section">
    <h2>About Me</h2>
    <p>Hello! I am learning how to code webpages using HTML for content and CSS for styling. I
t's fun and easy to get started!</p>
  </section>

  <!-- Hobbies Section (Cards Layout) -->
  <section id="hobbies" class="content-section">
    <h2>My Hobbies</h2>
    <div class="card-container">
      <div class="card">
        <h3>Coding 🖥️</h3>
        <p>Building websites and learning how the internet works step-by-step.</p>
      </div>
      <div class="card">
        <h3>Reading 📖</h3>
        <p>Exploring new ideas, stories, and technology blogs in my free time.</p>
      </div>
      <div class="card">
        <h3>Gaming 🎮</h3>
        <p>Relaxing with fun multiplayer and strategy games with friends.</p>
      </div>
    </div>
  </section>

  <!-- Simple Contact Form -->
  <section id="contact" class="content-section">
    <h2>Contact Me</h2>
```

```
<form>
  <label for="name">Your Name:</label>
  <input type="text" id="name" placeholder="Enter your name" required>

  <label for="email">Your Email:</label>
  <input type="email" id="email" placeholder="Enter your email" required>

  <label for="message">Message:</label>
  <textarea id="message" rows="4" placeholder="Write your message here..."></textarea>

  <button type="submit">Send Message</button>
</form>
</section>

<!-- Footer -->
<footer>
  <p>&copy; 2026 My Website. Created for learning HTML & CSS.</p>
</footer>

</body>
</html>
```

style.css - Part 1 (Lines 1-90)

```
/* =====
1. Basic Page Reset & Typography
===== */
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: Arial, sans-serif;
  line-height: 1.6;
  color: #333333;
  background-color: #f4f7f6;
}

/* =====
2. Header & Navigation Bar
===== */
header {
  background-color: #2c3e50;
  color: #ffffff;
  padding: 1rem 2rem;
  display: flex;
  justify-content: space-between;
  align-items: center;
}

header .logo {
  font-size: 1.5rem;
  font-weight: bold;
}

nav a {
  color: #ffffff;
  text-decoration: none;
  margin-left: 1.5rem;
  font-weight: 500;
}

nav a:hover {
  color: #1abc9c; /* Highlight color on hover */
}

/* =====
3. Hero Banner Section
===== */
.hero {
  background-color: #34495e;
  color: #ffffff;
  text-align: center;
  padding: 4rem 1rem;
}

.hero h1 {
  font-size: 2.5rem;
  margin-bottom: 0.5rem;
}
```

```
}

.hero p {
  font-size: 1.1rem;
  margin-bottom: 1.5rem;
  color: #ecf0f1;
}

.btn {
  display: inline-block;
  background-color: #1abc9c;
  color: #ffffff;
  padding: 0.75rem 1.5rem;
  text-decoration: none;
  border-radius: 4px;
  font-weight: bold;
}

.btn:hover {
  background-color: #16a085;
}

/* =====
   4. Main Content Sections
   ===== */
.content-section {
  max-width: 900px;
  margin: 2rem auto;
  padding: 1.5rem;
  background-color: #ffffff;
  border-radius: 8px;
  box-shadow: 0 2px 5px rgba(0, 0, 0, 0.1);
}
```

style.css - Part 2 (Lines 91-172)

```
.content-section h2 {
  color: #2c3e50;
  margin-bottom: 1rem;
  border-bottom: 2px solid #1abc9c;
  padding-bottom: 0.5rem;
}

/* =====
   5. Hobby Cards Layout
   ===== */
.card-container {
  display: flex;
  gap: 1rem;
  flex-wrap: wrap; /* Allows cards to wrap on smaller screens */
}

.card {
  flex: 1;
  min-width: 200px;
  background-color: #f8f9fa;
  padding: 1.25rem;
  border-radius: 6px;
  border: 1px solid #e9ecef;
}

.card h3 {
  margin-bottom: 0.5rem;
  color: #34495e;
}

/* =====
   6. Contact Form Styling
   ===== */
form {
  display: flex;
  flex-direction: column;
  gap: 0.75rem;
}

label {
  font-weight: bold;
  font-size: 0.9rem;
}

input, textarea {
  padding: 0.75rem;
  border: 1px solid #cccccc;
  border-radius: 4px;
  font-family: inherit;
}

input:focus, textarea:focus {
  outline: none;
  border-color: #1abc9c;
}
```



```
button[type="submit"] {
  background-color: #2c3e50;
  color: #ffffff;
  padding: 0.75rem;
  border: none;
  border-radius: 4px;
  font-size: 1rem;
  cursor: pointer;
}

button[type="submit"]:hover {
  background-color: #1abc9c;
}

/* =====
7. Footer
===== */
footer {
  text-align: center;
  padding: 1.5rem;
  background-color: #2c3e50;
  color: #ffffff;
  margin-top: 3rem;
  font-size: 0.9rem;
}
```

7. CODE EXPLANATION

CSS / HTML Property	Explanation
<code>*</code> (Universal Selector)	Selects all elements to reset margins and padding, creating a consistent baseline.
<code>box-sizing: border-box</code>	Includes padding and border within the element's total calculated width and height.
<code>display: flex</code>	Establishes a flex container, enabling one-dimensional layout alignment along axes.
<code>justify-content</code>	Defines alignment along the main axis (e.g., space-between for header logo and nav).
<code>align-items</code>	Controls alignment of flex items along the cross axis (e.g., center for vertical centering).
<code>gap</code>	Specifies the spacing between flex items without requiring child margin overrides.
<code>flex-wrap: wrap</code>	Allows flex items to automatically wrap onto multiple lines on smaller viewports.
<code>:hover</code>	Pseudo-class triggered when the user hovers over an element (e.g. nav links and buttons).
<code>:focus</code>	Pseudo-class applied when an input receives focus, styling border color to turquoise #1abc9c.
<code>border-radius</code>	Rounds the corners of outer borders for buttons, cards, and input fields.
<code>box-shadow</code>	Adds soft depth and elevation shadows to content containers and cards.
<code>line-height: 1.6</code>	Sets relative line spacing for optimal typographic readability.
<code><link rel="stylesheet"></code>	Connects external CSS file into the HTML document head.

8. TEST CASES AND EXECUTION DATA

The following test suite was executed manually in Google Chrome to verify CSS styling, responsive layout, and form focus:

TC ID	Test Description	Input / Action	Expected Result	Status
TC-01	External Stylesheet Linkage	Load index.html in browser	Stylesheet loaded; background displays #f4f7f6 and typography matches Arial	Pass
TC-02	Navigation Bar Hover State	Hover over navigation links	Link text color changes to bright turquoise #1abc9c	Pass
TC-03	Hero Call-to-Action Hover	Hover over 'Learn More' button	Button background transitions smoothly to dark emerald #16a085	Pass
TC-04	Hobby Card Flex Layout	View Hobbies section	3 cards (Coding, Reading, Gaming) display horizontally with equal spacing	Pass
TC-05	Responsive Card Wrapping	Narrow the browser viewport	Hobby cards wrap onto subsequent lines cleanly without clipping	Pass
TC-06	Form Input Focus Styling	Click inside Name/Email inputs	Outline removed and border changes to turquoise #1abc9c	Pass

9. OUTPUT

Output: Rendered Webpage in Chrome Browser

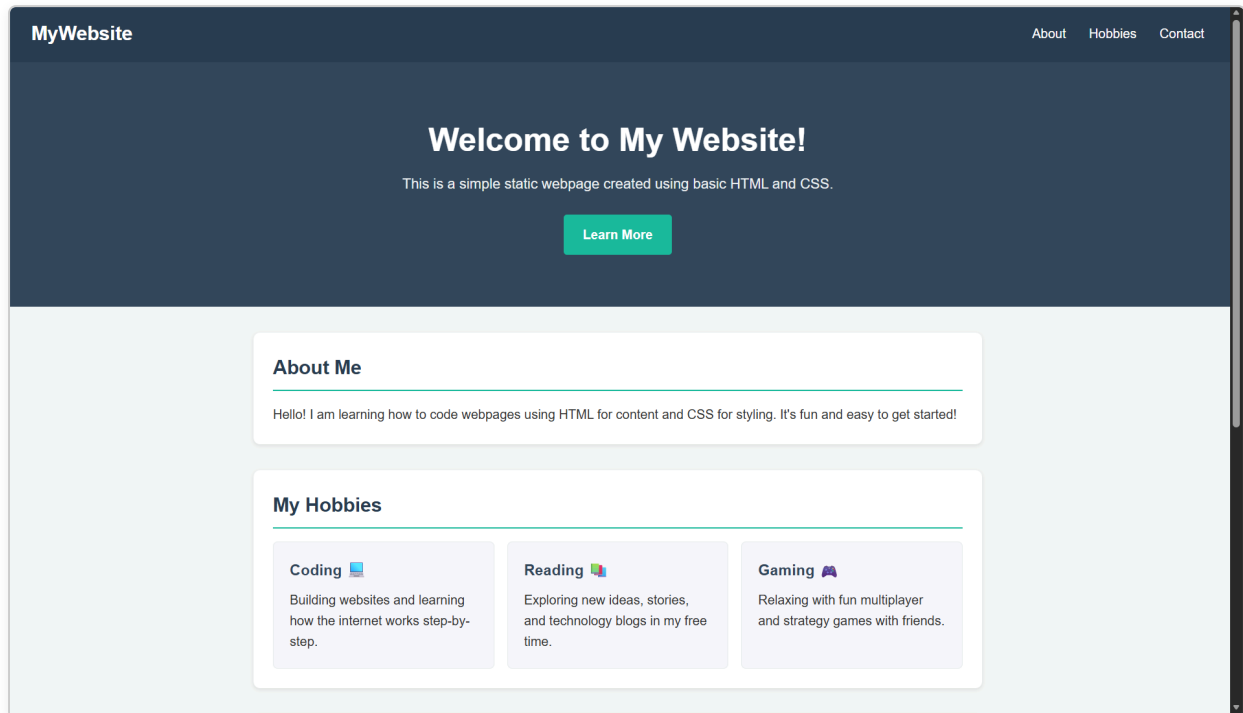


Figure 1: Styled webpage displaying responsive flex header, hero banner, hobby cards, and contact form.

10. RESULT

Thus, a modern webpage utilizing semantic HTML and external CSS was successfully designed, styled, and verified using manual test cases under browser defaults.

Submission Package Structure:

```
Experiment - 2(week -2)/
├── Documentation/
│   ├── Experiment_2.docx
│   └── Experiment_2.pdf
├── Screenshot/
│   └── Screenshot 2026-08-16 222914.png
└── Src/
    ├── index.html
    └── style.css
```